Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
6	Indicative content • Pyrophosphatase has tertiary structure • The active site has a specific shape • Pyrophosphate has a complementary shape • fits into the active site/ forms Enzyme Substrate Complex • lock and key theory/ induced fit • Phenylalanine must be a non-competitive inhibitor; • As it has a shape different to pyrophosphate; • Phenylalanine bonds to the allosteric site/site away from the active site; • Causes a change in the shape of the active site; • Pyrophosphate's active site is no longer complementary/prevents pyrophosphate binding • fewer enzyme-substrate complexes are formed; • The maximum rate of reaction cannot be reached at any concentration of pyrophosphate; • Phosphate must be a competitive inhibitor; • As it has a similar shape to {pyrophosphate/ substrate} / phosphate has a complementary shape the active site; • Phosphate {binds to/ competes for} the active site; • Prevents pyrophosphate binding / fewer enzyme-substrate complexes are formed; • The maximum rate of reaction can be reached at higher concentrations of pyrophosphate;						